

Fabio Battaglia

LEAD SOFTWARE ENGINEER ·

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Work Experience

Spire Global

Luxembourg, Luxembourg

SENIOR SOFTWARE ENGINEER

Dec. 2022 - Present

- Develop, deploy, and maintain core services for the Space Services unit, enabling effective management of custom satellite constellations.
- Led and delivered a 2-year development project co-funded by the European Space Agency (ESA), driving design, implementation, and integration phases.
- Architected microservices using **FastAPI, Flask, PostgreSQL, Terraform, Docker, Kubernetes, GitHub CI/CD actions**, minimizing manual deployments and accelerating release cycles.
- Collaborated with cross-functional teams—mission operators, hardware engineers, and product managers—to define requirements, establish timelines, and ensure successful project outcomes.
- Established robust monitoring and alerting solutions (e.g., Prometheus, Grafana), improving platform reliability and reducing mean time to recovery (MTTR).

Remark AI

London, United Kingdom

LEAD SOFTWARE ENGINEER

Sep. 2020 - Oct. 2022

- Architected and developed an application for thermal scanning, mask detection, and **face recognition** using thermal or standard network cameras, improving on-site health and security measures.
- Created microservices leveraging **Flask, FastAPI, C++, MongoDB, MySQL, GraphQL, Kafka, Traefik, Docker, Kubernetes** and deployed them via GitLab CI/CD (on AWS and on-premise).
- Implemented and optimized AI pipelines (e.g., **DeepStream** in C++/Python) for real-time face recognition, collaborating with data scientists to refine model accuracy and speed.
- Onboarded and mentored backend and frontend engineers, including code reviews, pair programming, and knowledge-sharing sessions.
- Coordinated sprint planning and Agile ceremonies (Scrum/Kanban), fostering team alignment and consistently delivering on schedule.

Amadeus

London, UK

SENIOR SOFTWARE ENGINEER

Sep. 2017 - Sep. 2020

- Enhanced Amadeus Altéa DCS (C++11), coordinating urgent deliveries (China authorities submission).
- Prototyped a rules-based system (Drools, Java, MongoDB) and mentored junior developers.

XiD Technologies

Singapore

SOFTWARE ENGINEER

Feb. 2015 - Aug. 2017

- Integrated 3rd-party biometric devices (face/fingerprint) into XiD FaaS and built embedded solutions.
- Implemented security systems (CCTV, fingerprint/RFID) for Changi Airport T5, enabling real-time alerts.

ENEA

Rome, Italy

CONTROL SYSTEMS ENGINEER (INTERN)

Jun. 2014 - Oct. 2014

- Developed a real-time monitoring tool (MATLAB/C++) for Tokamak thermal behavior in nuclear fusion research.

High School C. Marchesi

Catania, Italy

ROBOTICS TEACHER

Mar. 2014 - Jun. 2014

- Taught an EU-PON-funded robotics course for 20 students, covering theory and hands-on projects (Arduino, sensors).

Education

University of Catania

Catania, Italy

MASTER'S IN AUTOMATION ENGINEERING & CONTROL OF COMPLEX SYSTEMS

Apr. 2012 - Oct. 2014

- Focused on Control Systems, Robotics, Image Processing, & Industrial Automation.
- Thesis on Tokamak thermal phenomena (ESA research). Graduated with **110/110 cum laude**.

University of Catania

Catania, Italy

BACHELOR'S IN COMPUTER ENGINEERING

Oct. 2008 - Mar. 2012

- Studied core Engineering (Math, Electronics, Networking, OS, C/C++/Java).
- Thesis: "NetFPGA Architecture & Hardware Description."

Skills

Programming Languages

C++11+, Python 3, TypeScript, Rust, C, C#, Java, Matlab, $\LaTeX$

Python Ecosystem

FastAPI, Flask, Pydantic, SQLAlchemy, PyTest, MyPy

DevOps & Cloud

Docker, Kubernetes, Terraform, Jenkins, GitLab/GitHub CI/CD, AWS (S3, EKS, Lambda, RDS)

Databases

PostgreSQL, MySQL, MSSQL, Elasticsearch, MongoDB

Electronics & Embedded

STM32, Arduino, Raspberry Pi, PCB design, Siemens PLCs